

LECTURE 8

Fluid Kinematics

Reference: Munson, Rothmayer, Okiishi & Huebsch, Fundamentals of Fluid Mechanics, Chapter 4
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Fluid Kinematics

Kinematics is the description of motion.

Here we will describe how a fluid moves — its velocity, its acceleration, and the lines that make the motion visible — without asking what forces produce the motion.

The velocity field, and the two ways of describing it

Streamlines, streaklines and pathlines

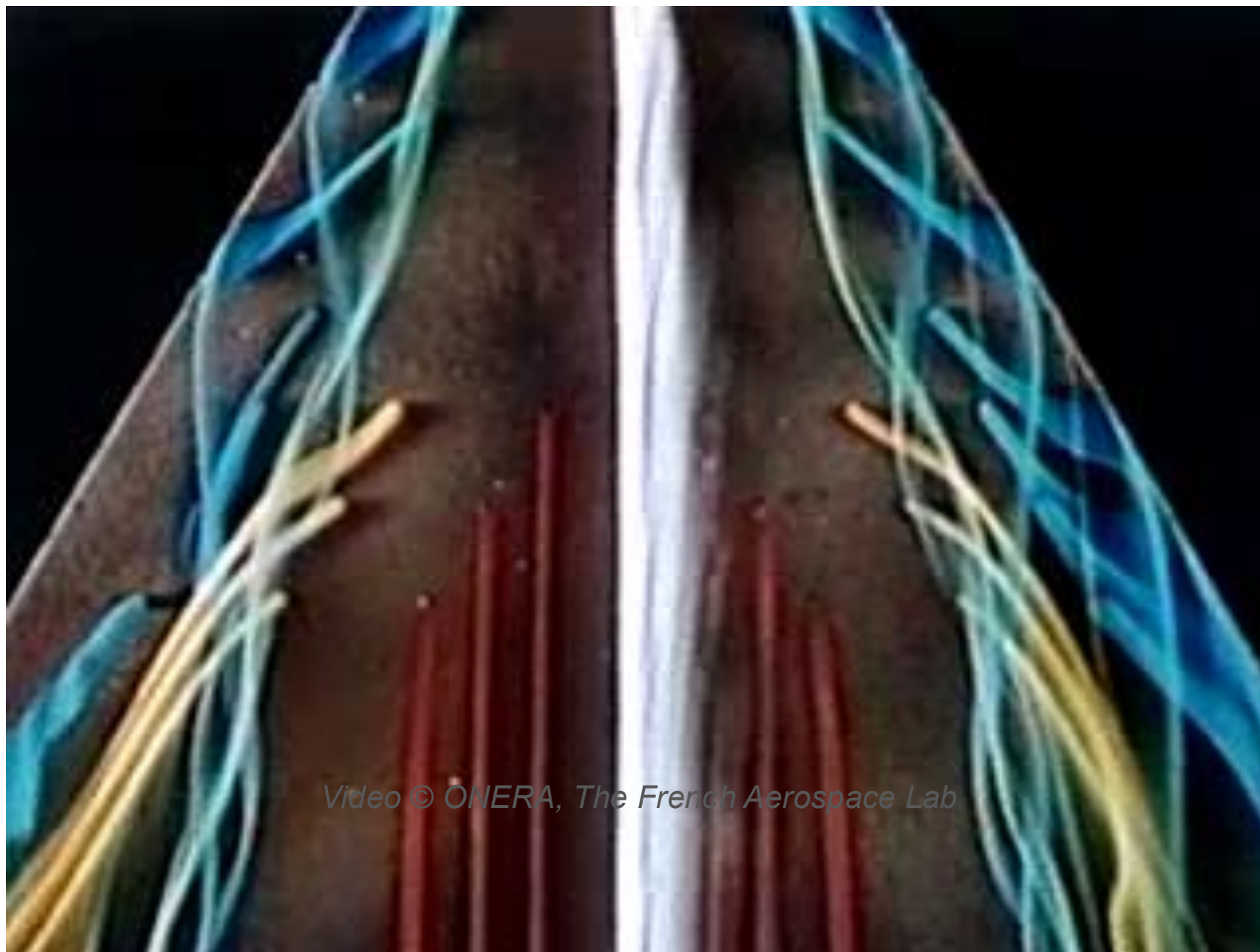
The acceleration field and the material derivative

Systems, control volumes and the Reynolds transport theorem

Streaklines past a delta wing

Flow past a delta wing model, made visible by injecting dyes of several colours.

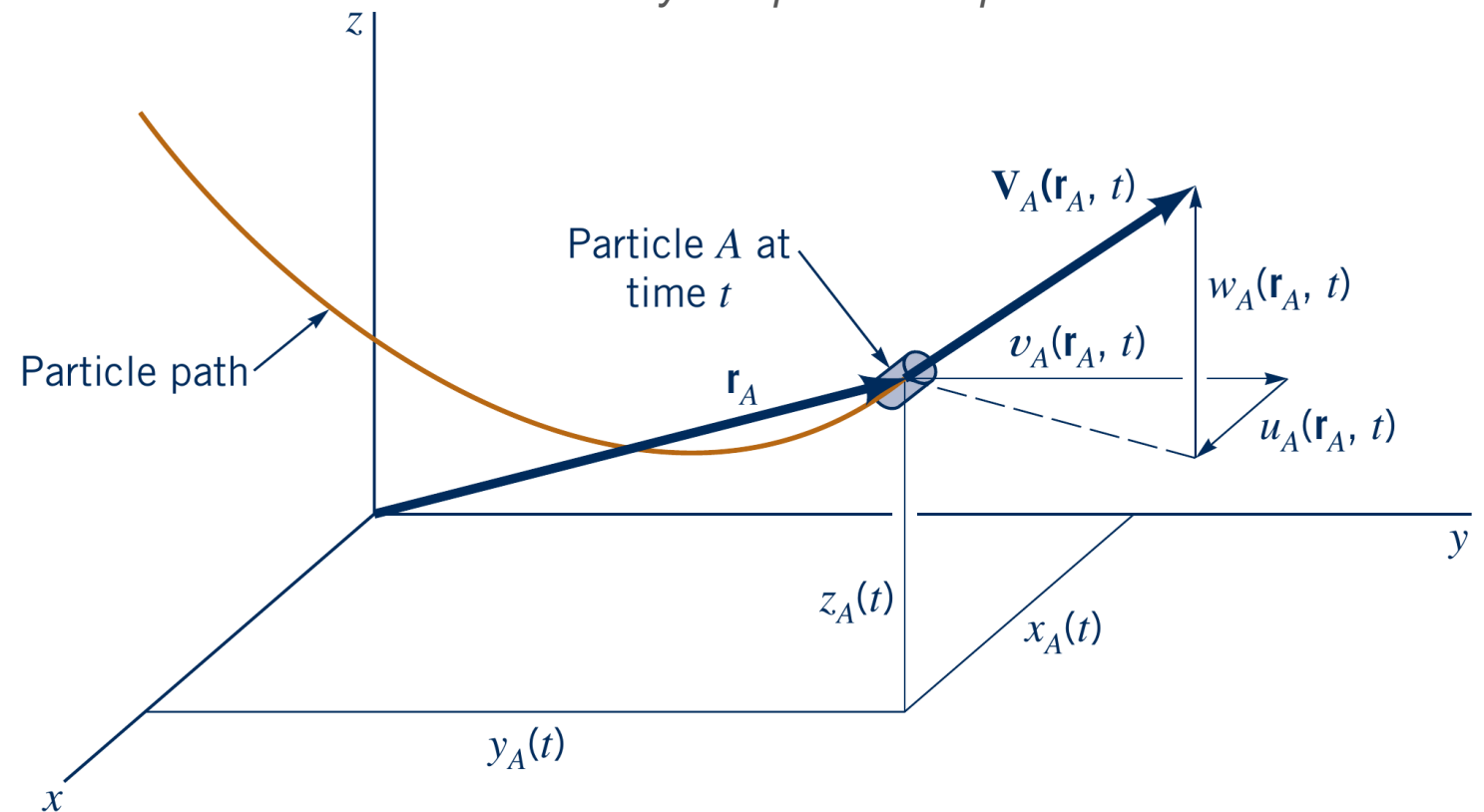
The dyed fluid forms lines called **streaklines**, and the structure and shape of those lines carry a great deal of information about the flow.



Video © ONERA, The French Aerospace Lab

The velocity field

Velocity and position of particle A at time t . Munson.



A small volume of fluid holds far too many molecules to follow one at a time.

We use the continuum hypothesis and work with fluid particles instead, giving every property a value at every point at every instant — **a field**.

$$\mathbf{v} = u(x, y, z, t) \hat{i} + v(x, y, z, t) \hat{j} + w(x, y, z, t) \hat{k}$$

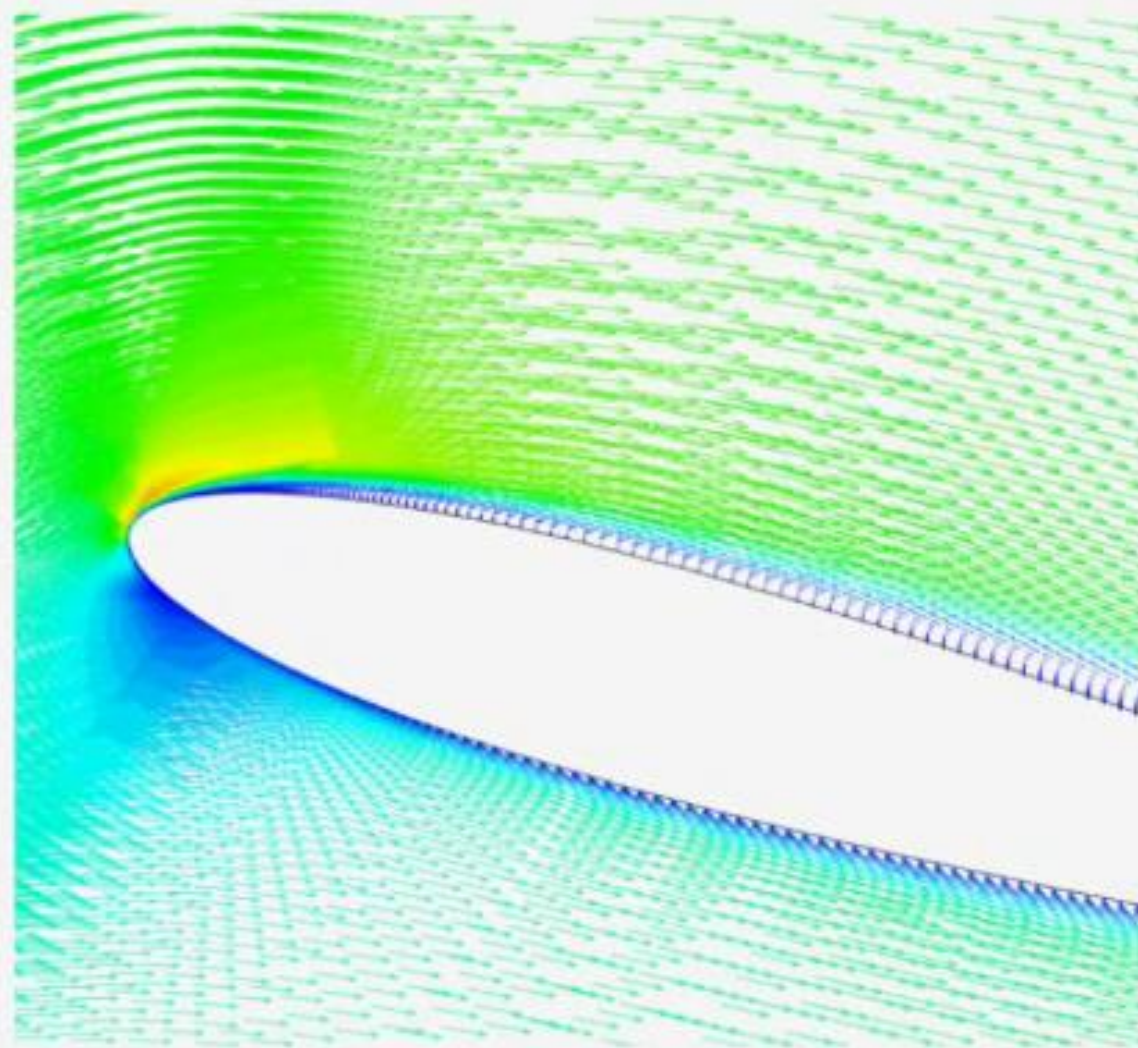
Here u , v and w are the velocity components, and the speed is the magnitude of the vector.

$$|\mathbf{v}| = (u^2 + v^2 + w^2)^{1/2}, \quad \frac{d\mathbf{r}_A}{dt} = \mathbf{v}_A$$

The velocity field about an airfoil

The calculated velocity field about the front of an airfoil as the angle of attack increases — the angle the airfoil makes with the oncoming stream.

The length and colour of each arrow give the fluid speed; the direction of the arrow gives the instantaneous flow direction.

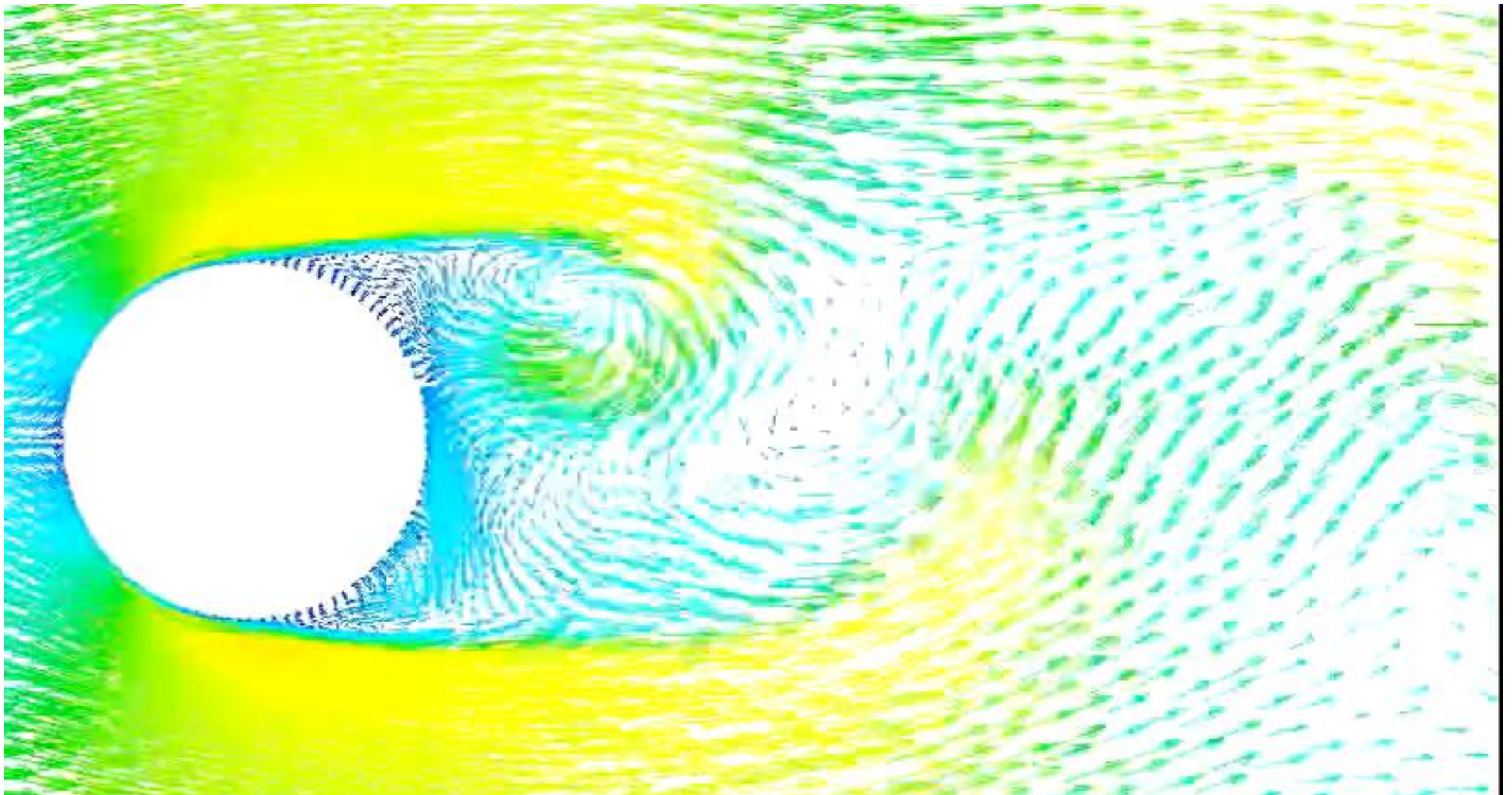


Video courtesy of Chris Griffin, West Virginia University

Velocity vectors past a cylinder

The unsteady, oscillating velocity field past a circular cylinder.

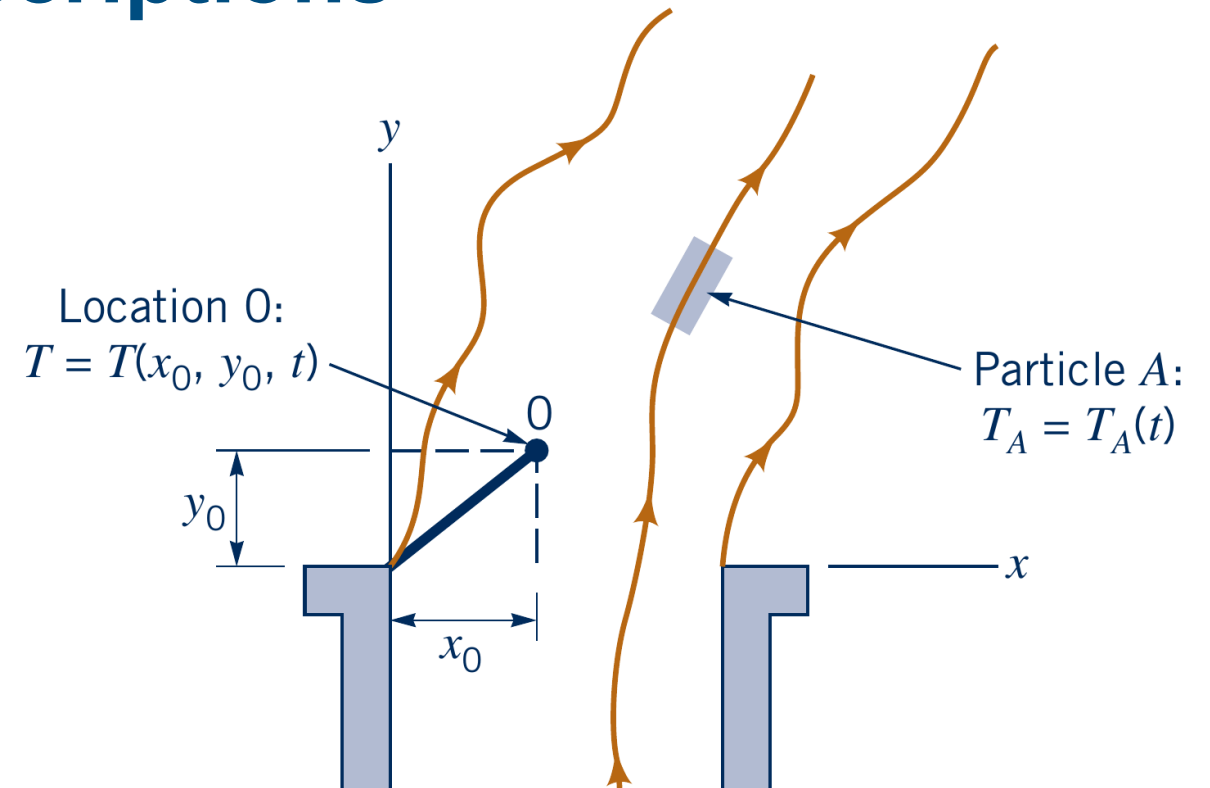
The upstream flow is steady, yet the motion behind the cylinder is a regular train of shed vortices — **the Kármán vortex street**.



Video © ANSYS, Inc.

Eulerian and Lagrangian descriptions

There are 2 ways of describing the same flow:



Eulerian and Lagrangian descriptions of the temperature of a flowing fluid.

Eulerian:

Fix your instrument in space and record what passes it. You obtain the field, $T = T(x, y, z, t)$. You never need to know which particle you measured.

Lagrangian:

Tag a particle and follow it. You obtain that particle's history, $T_a = T_a(t)$. You never need to know what happens anywhere else.

Both contain the same information. In fluid mechanics the Eulerian description is easier, which is why the field representation is the one we use.

Following the particles: experiment

Convection inside a small water droplet on a flat surface, with the particles lit by a laser sheet.

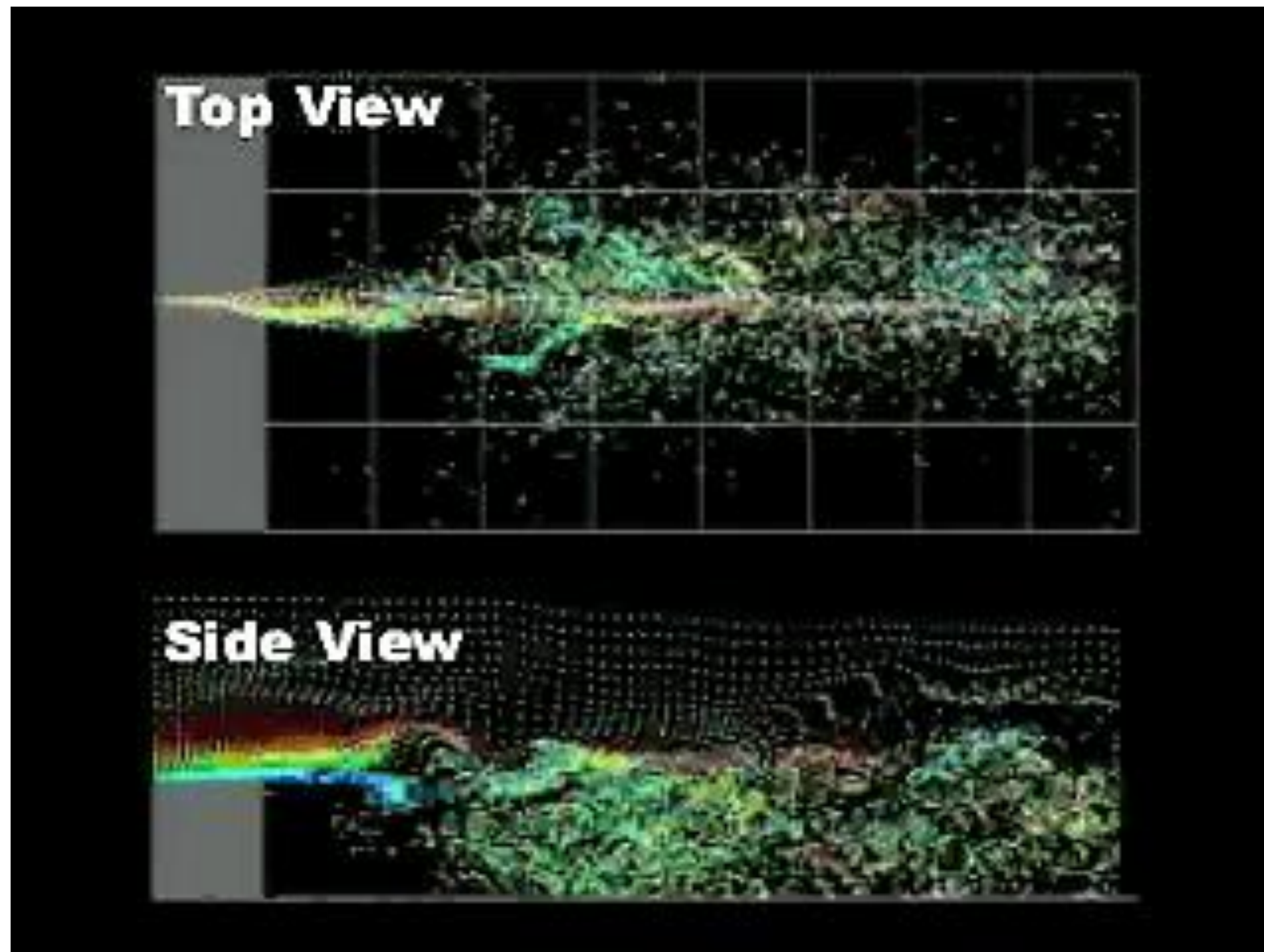
Following one particle illustrates the Lagrangian description. Note also that some particles move out of the plane of view, so this flow is 3D.



Following the particles: computation

A computed 3D flow past a backward-facing step.

Small particles are inserted numerically and followed as the field carries them, which turns an Eulerian solution into a Lagrangian picture of the same flow.



Dimensionality and steadiness

Dimensionality: a real flow is 3D and time dependent, but very often 1 or 2 components are small enough to drop:

$$\mathbf{V} = u \hat{i} \qquad \mathbf{V} = u \hat{i} + v \hat{j} \qquad \mathbf{V} = u \hat{i} + v \hat{j} + w \hat{k}$$

Almost no flow is truly 1D. The approximation is good in pipes and ducts, and poor wherever the flow separates or swirls.

Steadiness: a flow is steady when nothing at a fixed point in space changes with time:

$$\left. \frac{\partial \mathbf{V}}{\partial t} \right|_{x,y,z} = 0$$

This is a statement about a point, not about a particle.

In steady flow a particle's own properties can still change as it moves: the exhaust leaving a pipe at a constant temperature cools as it travels away.

Unsteady flows may be non-periodic — a tap being shut; **periodic** — the charge entering an engine cylinder; or **random**, which is what turbulence is.

Flow past a wing

A model Airbus flown through two plumes of smoke.

The complex, unsteady, 3D swirl shed at the wing tips — **the trailing vortices** — is clearly visible. Vorticity and turbulence are 3D.



Types of flow: laminar vs. turbulent

Video © B. R. Munson



Low-speed water from a small nozzle is **steady and laminar**: unless you disturb it, you cannot tell the fluid is moving at all.

The flow inside a clothes washer is the opposite — **highly unsteady and turbulent**, which is what does the cleaning.

Streamlines

A streamline is a line that is everywhere tangent to the velocity field. In 2D, its slope is fixed by the ratio of the two velocity components:

$$\frac{dy}{dx} = \frac{v}{u}$$

Integrating this equation gives the family of streamlines.

- **If the flow is steady**, the velocity direction at every point never changes, so the streamlines are fixed curves in space.
- **If the flow is unsteady**, they change shape from instant to instant, and there is no straightforward way to produce them in the laboratory.

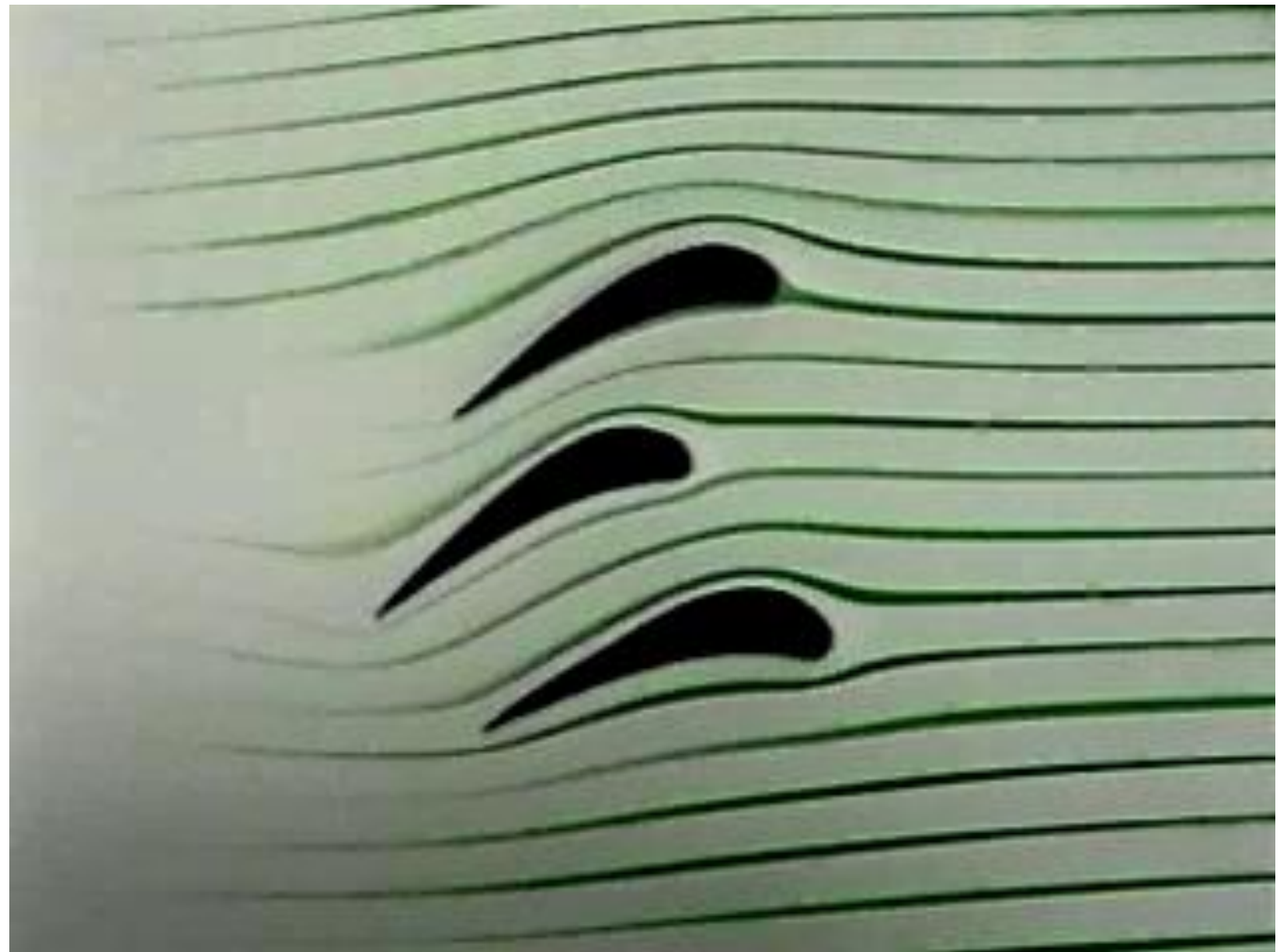
A streamline is an analytical object:

You get it by integrating the field, not by photographing the flow.

Streamlines around cylinders and airfoils

Dye injected into water flowing steadily around a series of cylinders.

Because the flow is steady, these lines are fixed in space and show the paths the particles follow.



Video © B. R. Munson

On the airfoils, the stagnation streamline and its stagnation point can be read straight off the picture.

Streaklines and pathlines

Streakline:

all the particles that have already passed through one fixed point. This is what you see when dye or smoke is injected continuously at one place.

Pathline:

the line traced out by one particle as it travels. A Lagrangian object; in the laboratory, a time exposure of a single marked particle.

- In steady flow the streamline, the streakline and the pathline through a given point are the same line.
- In unsteady flow none of the three need to coincide.

Photographs captioned “streamlines” almost always show streaklines.

- For steady flow, the distinction does not matter.
- For unsteady flow it matters a great deal, and the next example shows how far apart the three can be.

Streaklines past a flat plate

Smoke injected continuously upstream of a flat plate held normal to the flow.

The geometry is simple and the upstream flow is steady, yet the resulting flow is unsteady — and because of that, these streaklines are not the streamlines.



Pathlines from a sprinkler

A lawn sprinkler. Once the water leaves the arm nothing acts on it, so each particle travels a nearly straight radial line: those are the pathlines.

The visible spiral of the water streams is not a pathline.



Jupiter's red spot

The great red spot is a vortex about 7 times the diameter of the Earth. The turbulent, eddying motion in this Voyager footage is made visible by the natural colour contrast of the atmosphere: no dye or smoke needed.



Video courtesy of NASA

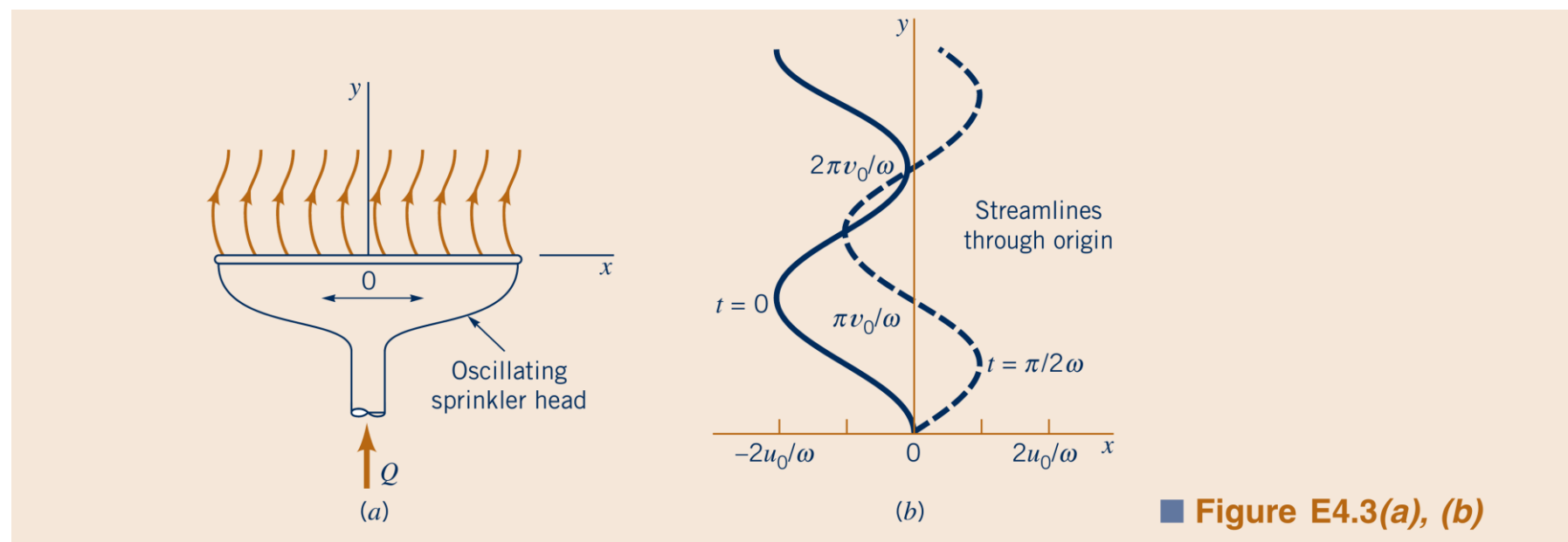
Example: the three lines compared

Water leaves a slit that oscillates from side to side. The vertical velocity is uniform, $v = v_0$, and the transverse velocity is carried upward unchanged:

$$\mathbf{v} = u_0 \sin \left[\omega \left(t - \frac{y}{v_0} \right) \right] \hat{i} + v_0 \hat{j}$$

Streamlines: we freeze the clock, integrate $dy/dx = v/u$ at $t = 0$ and $t = \pi/2\omega$:

$$x = \frac{u_0}{\omega} \left[\cos \left(\frac{\omega y}{v_0} \right) - 1 \right] \quad \text{and} \quad x = \frac{u_0}{\omega} \sin \left(\frac{\omega y}{v_0} \right)$$



(a) the oscillating slit; (b) streamlines through the origin at $t = 0$ (solid) and at $t = \pi/2\omega$ (dashed): the pattern itself changes with time. Munson et al., Fig. E4.3(a), (b)

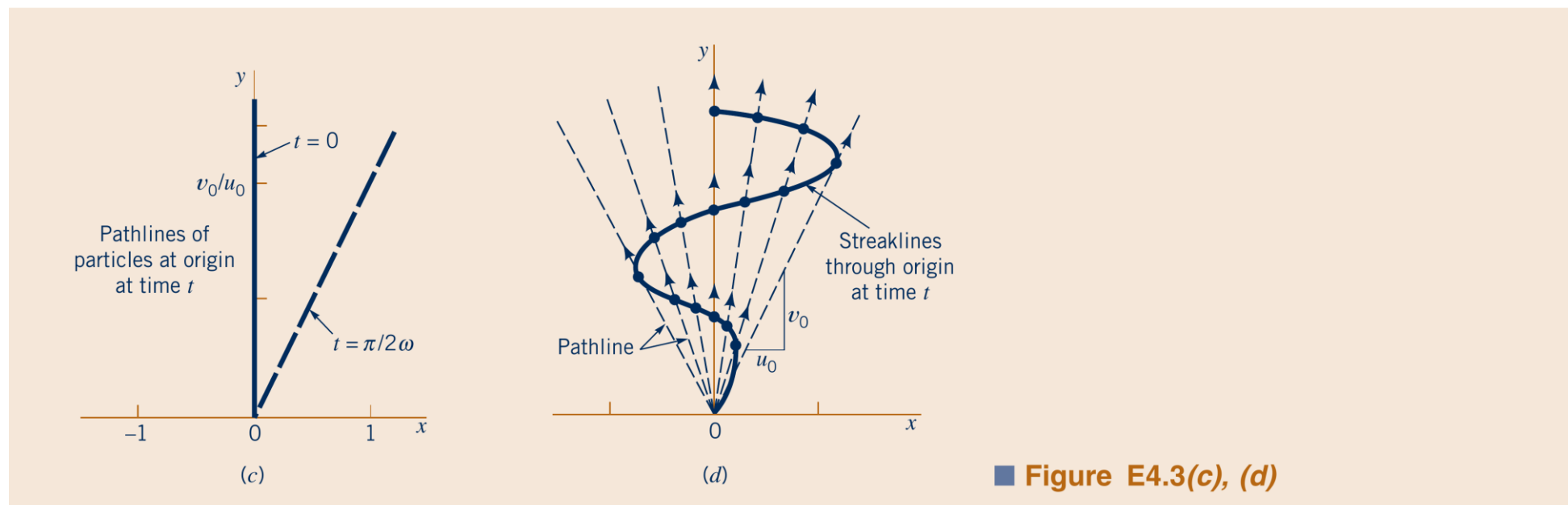
Pathlines and the streakline

Pathlines: we follow 1 particle. Since $y = v_0 t$, the phase $\omega(t - y/v_0)$ keeps its release value, so each particle runs along a straight ray:

$$x = 0, \quad y = v_0 t \quad \text{and} \quad y = \frac{v_0}{u_0} x$$

Streakline: the dye now at height y left the slit at $t - y/v_0$:

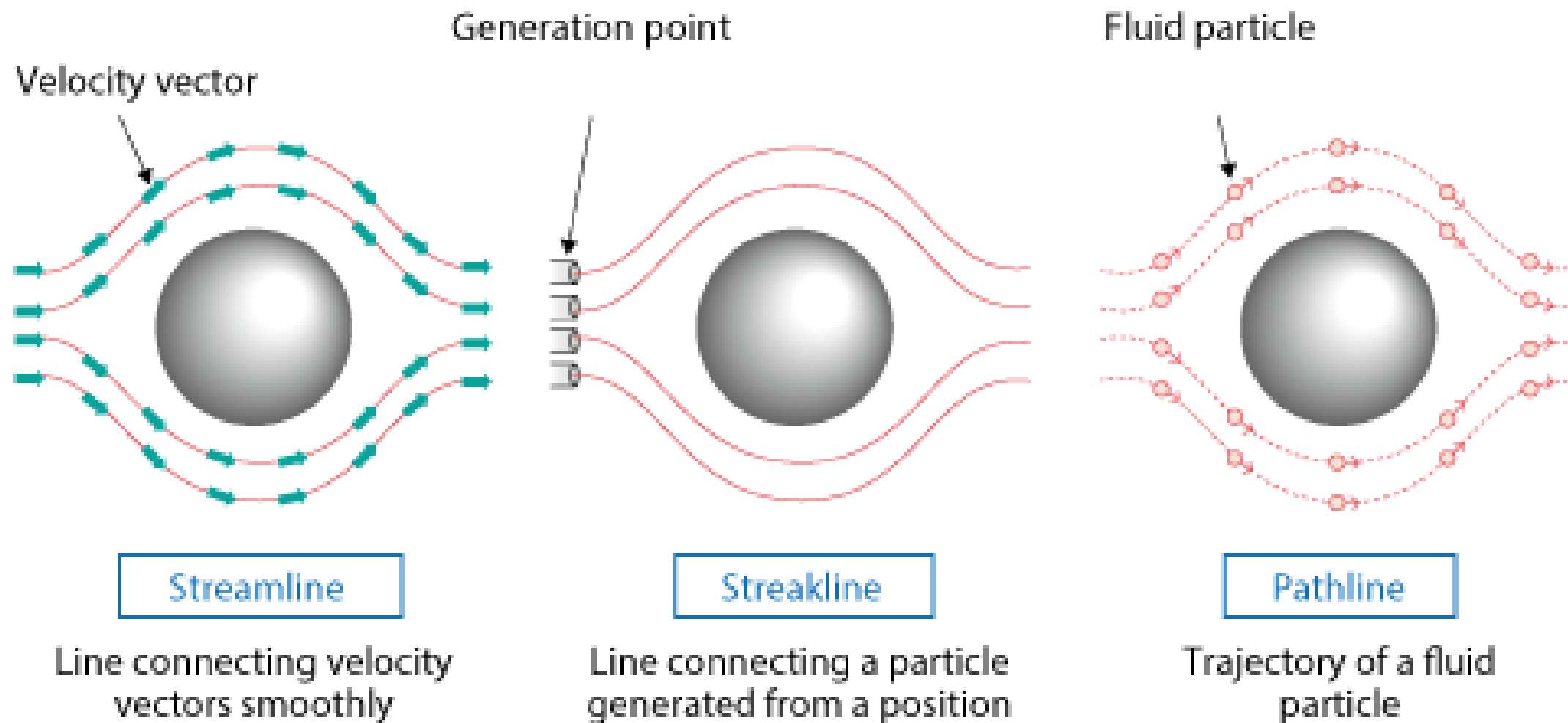
$$x = \frac{u_0 y}{v_0} \sin \left[\omega \left(t - \frac{y}{v_0} \right) \right]$$



(c) pathlines are straight rays from the origin; (d) the streakline is the sinuous curve that threads across them. Munson et al., Fig. E4.3(c), (d)

Summary: Streamline, streakline and pathline

Credits: <https://www.cradle-cfd.com/media/column/a147>



Streamline is a curve, which smoothly connects velocity vectors at a certain moment.

The tangent line of a streamline indicates the flow direction on the streamline.

A **streakline** is a curve, which is drawn by connecting particles generated from a certain point in a space.

A trace of smoke from a chimney is a typical example of a streakline.

A **pathline** is a trajectory of a fluid particle.

A trajectory of a balloon in a space is an example of a pathline.

The acceleration field

Newton's second law needs the acceleration of a particle, but we have chosen to describe the flow with a field.

We need the chain rule: a particle's velocity changes both because time passes and because the particle has moved somewhere else.

$$\mathbf{a} = \frac{\partial \mathbf{v}}{\partial t} + u \frac{\partial \mathbf{v}}{\partial x} + v \frac{\partial \mathbf{v}}{\partial y} + w \frac{\partial \mathbf{v}}{\partial z}$$

In components,

$$a_x = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z}$$

with the same form for the y and z components

Nothing here requires us to know which particle we are talking about:
the acceleration is a field in its own right.

The material derivative

The combination that appeared in the acceleration is a differential operator: **the material derivative (total derivative)**

$$\mathbf{a} = \frac{D\mathbf{v}}{Dt}, \quad \frac{D(\quad)}{Dt} \equiv \frac{\partial(\quad)}{\partial t} + (\mathbf{v} \cdot \nabla)(\quad)$$

It gives the rate of change of any quantity for a particle moving with the fluid, computed entirely from field information.

Applied to temperature, for example,

$$\frac{DT}{Dt} = \frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T$$

Nothing about it is special to fluid mechanics. All it needs is a field $T(x, y, z, t)$ and the velocity of the particle that moves through it.

Local and convective effects

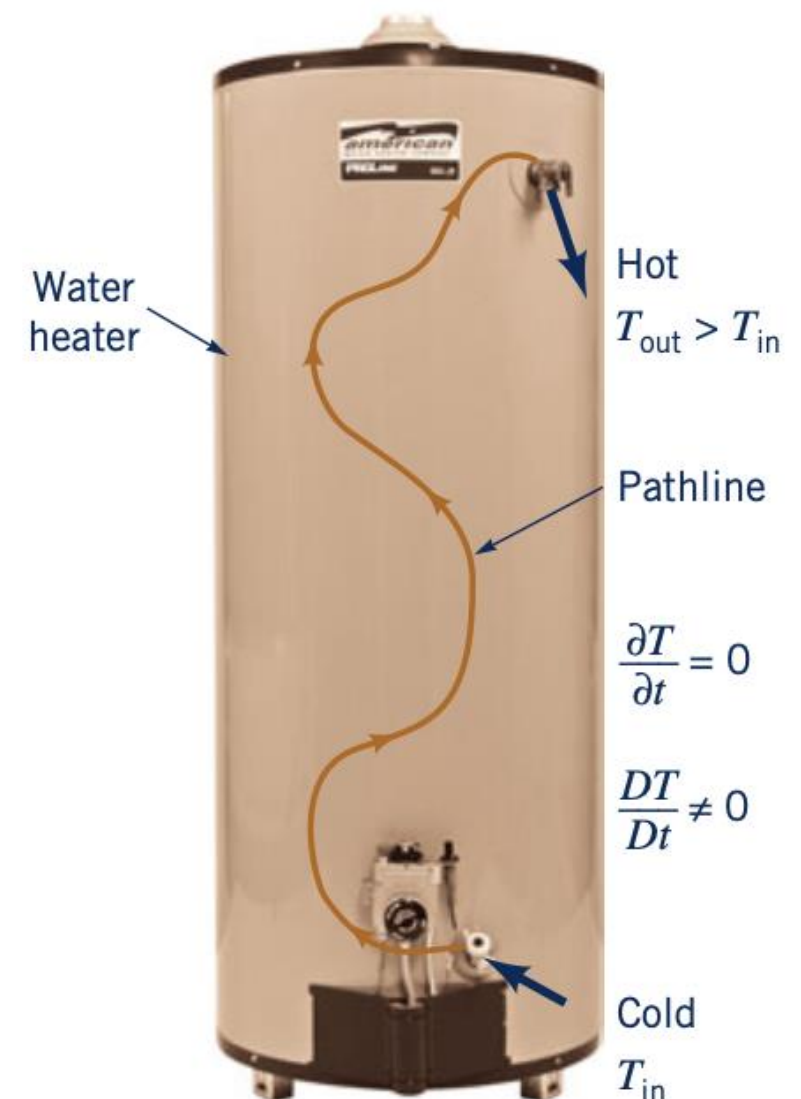
The material derivative always splits into 2 parts, and they have different physical origins:

$$\mathbf{a} = \underbrace{\frac{\partial \mathbf{v}}{\partial t}}_{\text{local}} + \underbrace{(\mathbf{v} \cdot \nabla) \mathbf{v}}_{\text{convective}}$$

The **local term** is the unsteadiness of the flow and vanishes in steady flow.

The **convective term** is the effect of the particle arriving where the quantity has a different value and survives even in steady flow.

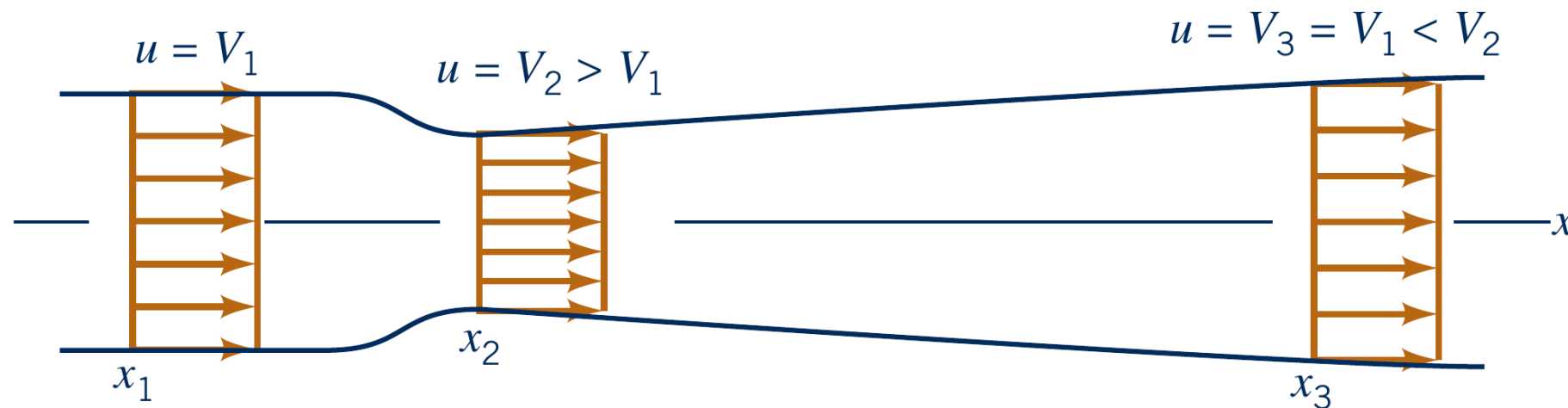
A water heater is an example. The flow is steady, so the temperature at any fixed point never changes, $\partial T / \partial t = 0$. Yet every particle enters cold and leaves hot: $DT/Dt \neq 0$, entirely through the convective term.



Local and convective effects

$$\mathbf{a} = \underbrace{\frac{\partial \mathbf{v}}{\partial t}}_{\text{local}} + \underbrace{(\mathbf{v} \cdot \nabla) \mathbf{v}}_{\text{convective}}$$

Example: Convective acceleration in a variable-area pipe



Uniform, steady flow in a variable area pipe: the fluid accelerates between sections 1 and 2 with no local acceleration at all.

- Steady, 1D flow: $\partial v / \partial t = 0$
- Fluid particles can still accelerate due to spatial changes in velocity: $a_x = u \partial u / \partial x$
- **Converging section:** u increases $\rightarrow \partial u / \partial x > 0 \rightarrow a_x > 0$ (acceleration)
- **Diverging section:** u decreases $\rightarrow \partial u / \partial x < 0 \rightarrow a_x < 0$ (deceleration)
- If $v_1 = v_3$, the total acceleration and deceleration balance, even if the 2 sections have different lengths.

Example: the acceleration field

Given

Let us consider a steady, 2D flow (see notebook 3), with v_0 and ℓ constant:

$$\mathbf{v} = \frac{v_0}{\ell} (-x \hat{i} + y \hat{j}) \quad \Longrightarrow \quad u = -\frac{v_0 x}{\ell}, \quad v = \frac{v_0 y}{\ell}$$

Find

The acceleration field.

Solution

The flow is steady, so $\partial v / \partial t = 0$; it is 2D, so $w = 0$ and $\partial(\cdot) / \partial z = 0$.

Only 2 convective terms survive:

$$\begin{aligned} \mathbf{a} &= u \frac{\partial \mathbf{v}}{\partial x} + v \frac{\partial \mathbf{v}}{\partial y} \\ &= \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) \hat{i} + \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) \hat{j} \end{aligned}$$

What the acceleration field looks like

Carrying out the derivatives with $u = -v_0 x/\ell$ and $v = v_0 y/\ell$:

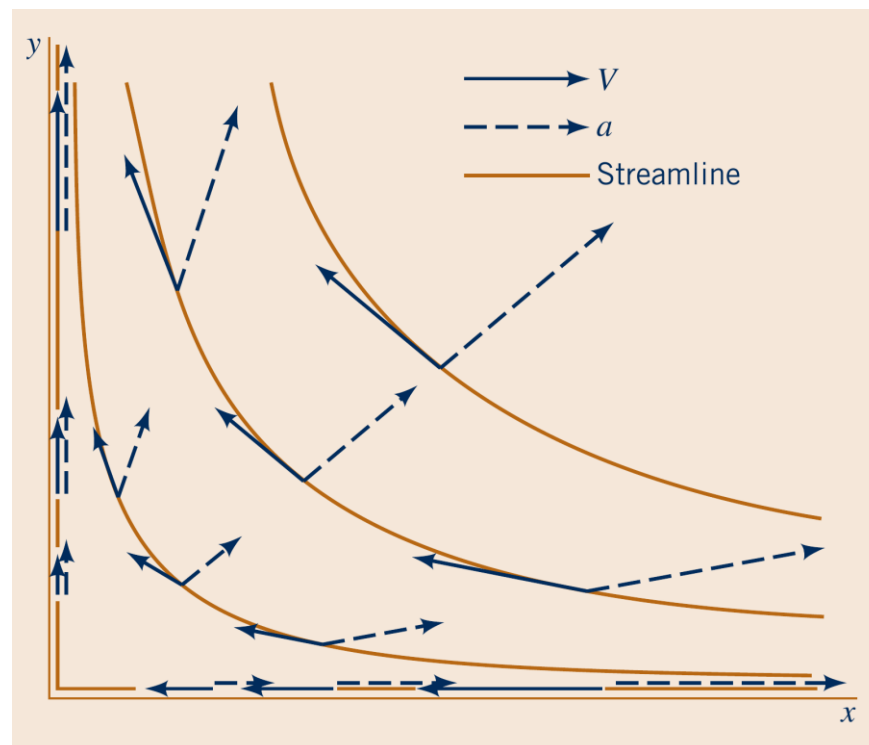
$$a_x = \frac{v_0^2 x}{\ell^2}, \quad a_y = \frac{v_0^2 y}{\ell^2}, \quad |\mathbf{a}| = \left(\frac{v_0}{\ell}\right)^2 (x^2 + y^2)^{1/2}$$

Direction: the components are in the ratio y/x , so they are rays from the origin.

Magnitude: $|a|$ grows with distance from the origin; a constant on circles.

Not parallel: v and a differ except on the axes, so pathlines curve.

At the origin: both vanish; a particle there stays put.

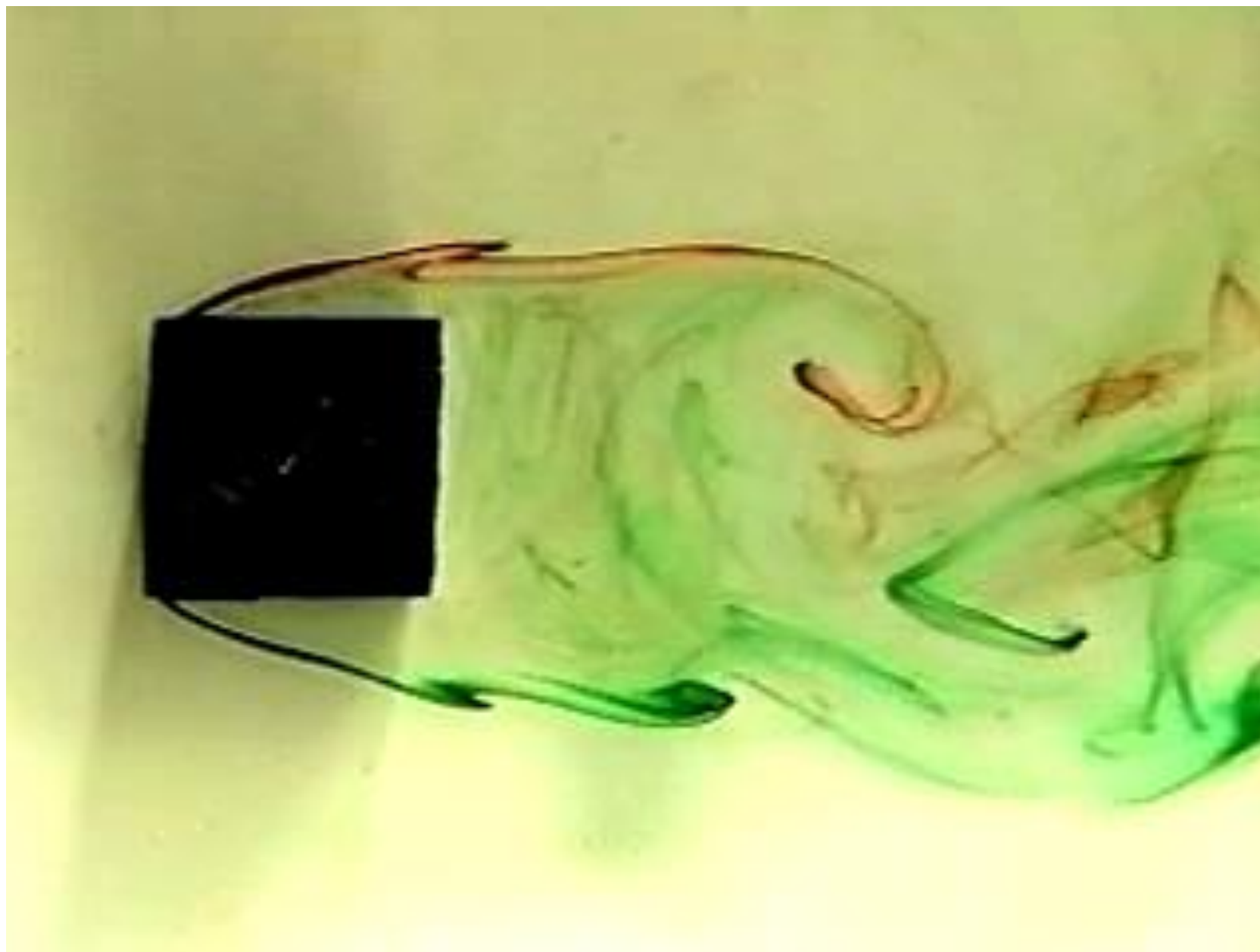


Velocity (solid) and acceleration (dashed) vectors in the first quadrant. Munson et al., Fig. E4.5

Unsteady flow

The upstream velocity is steady and the block is stationary, yet the flow is unsteady: vortices are shed alternately from the two sides of the block.

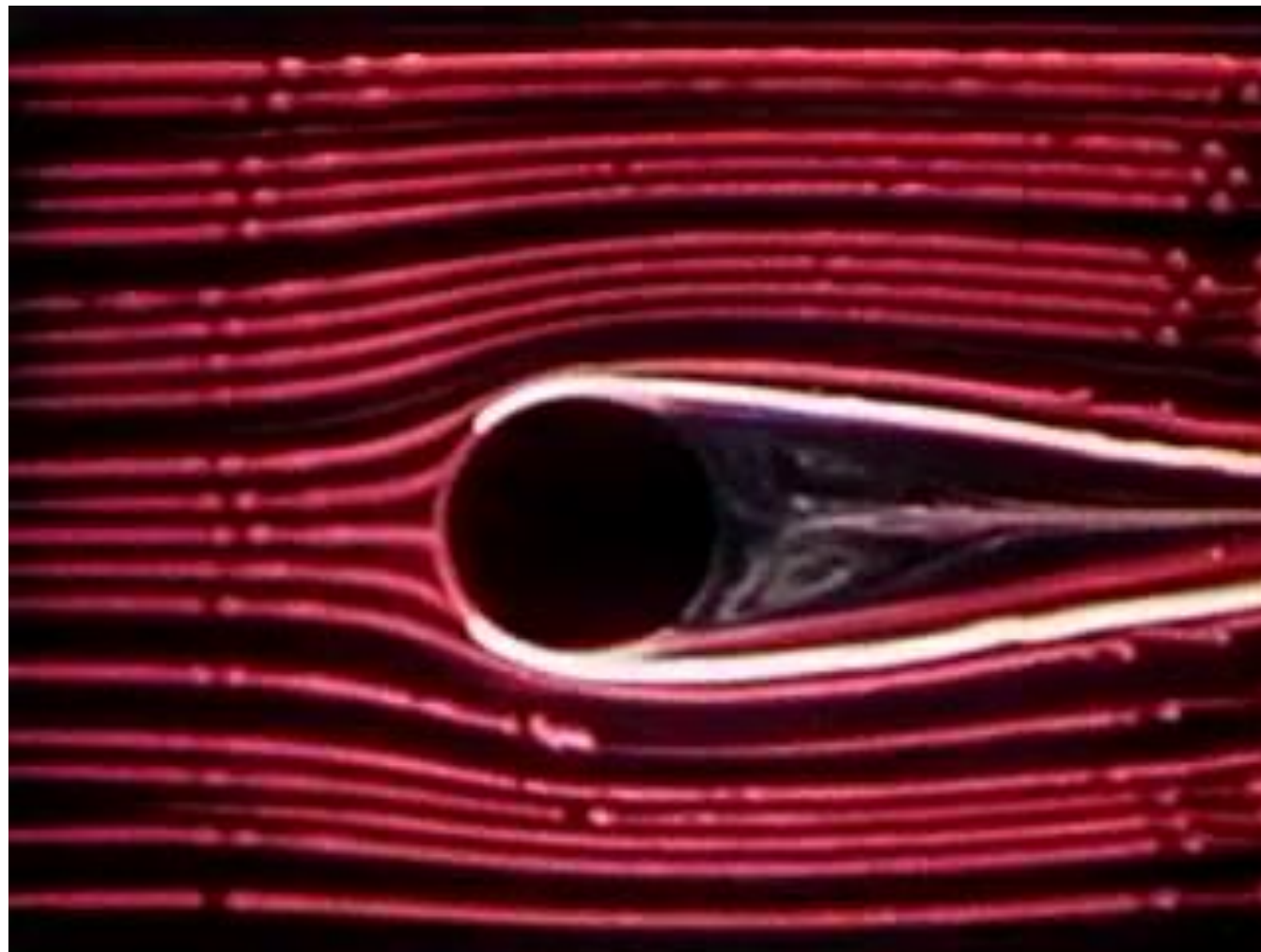
There is no obvious cause — the unsteadiness is generated by the flow itself.



Streamline coordinates

Steady flow past a circular cylinder, described in terms of the distance s measured along a streamline and the distance n normal to it

In front of and around the cylinder these coordinates are the natural ones; in the separated region behind it, far less so.



Video © ONERA, The French Aerospace Lab

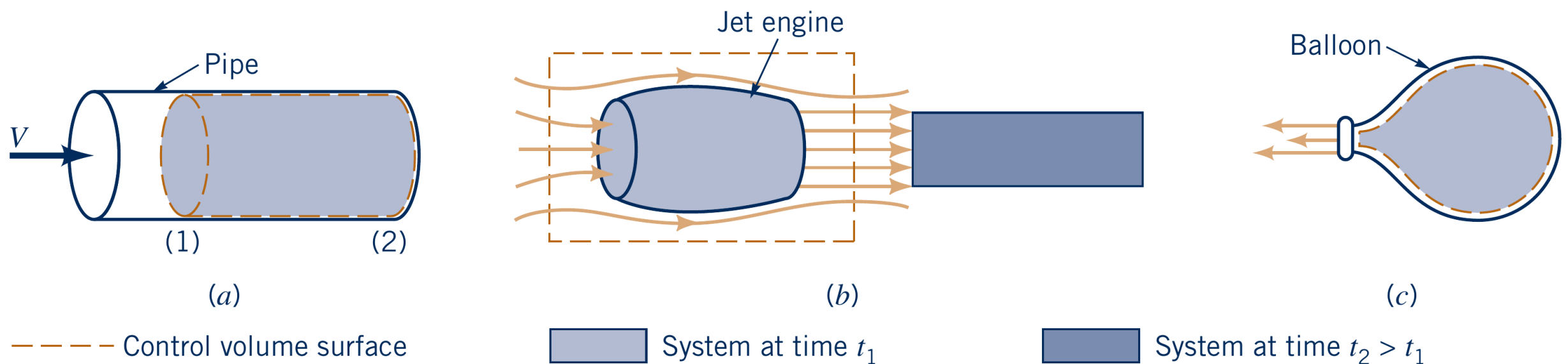
Systems and control volumes

System. a fixed quantity of matter — always the same particles. It may move, deform and change temperature, but its mass is fixed. **The Lagrangian object.**

Control volume. a region of space through which fluid flows. The matter inside it changes from instant to instant, and so may the amount of it. **The Eulerian object.**

The laws of mechanics are stated for systems:
the mass of a system is constant, the rate of change of its momentum equals the force on it.

But what we usually want is the thrust on an engine, not the history of the air that went through it. To use the laws on a control volume we must first rewrite them.



Fixed, moving and deforming control volumes.

Extensive and intensive properties

Let B stand for any property of the fluid — mass, momentum, energy, angular momentum — and let b be the amount of that property per unit mass:

$$B = m b, \quad B_{\text{sys}} = \int_{\text{sys}} \rho b \, d\mathcal{V}$$

B is extensive: it scales with how much matter you are considering.

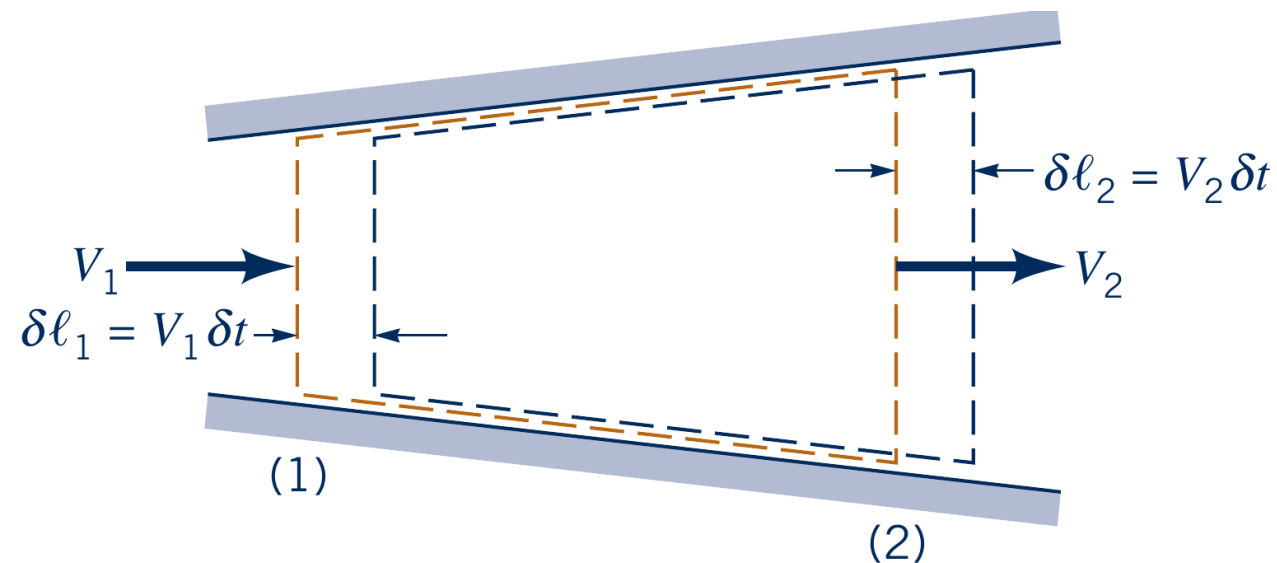
b is intensive: it does not

- If B is the mass, then $b = 1$.
- if B is the momentum, then b is the velocity
- if B is the kinetic energy, then $b = |v|^2/2$.

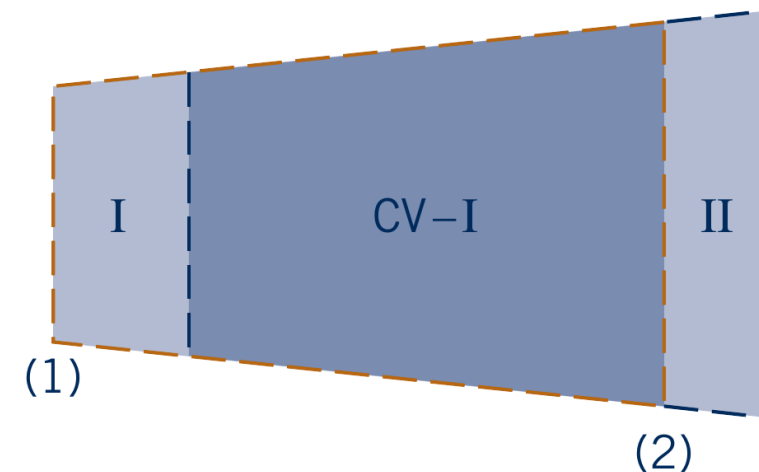
Almost every governing law is a statement about the rate of change of B for a system.

The **Reynolds transport theorem** turns that into quantities we can evaluate on a control volume.

Reynolds transport theorem



- — — Fixed control surface and system boundary at time t
- — — System boundary at time $t + \delta t$



Control volume and system for flow through a variable area pipe.

Take a fixed control volume inside a duct, and let the system be the fluid that occupies it at time t .

A moment later the system has moved: it has left the sliver I behind at the inlet and pushed into the sliver II beyond the outlet. Comparing the two instants and letting $\delta t \rightarrow 0$ gives

$$\frac{DB_{\text{sys}}}{Dt} = \frac{\partial B_{\text{cv}}}{\partial t} + \rho_2 A_2 v_2 b_2 - \rho_1 A_1 v_1 b_1$$

The rate of change for the system differs from the rate of change inside the control volume by exactly the net rate at which B is carried across the two faces.

The general form: Reynolds transport theorem

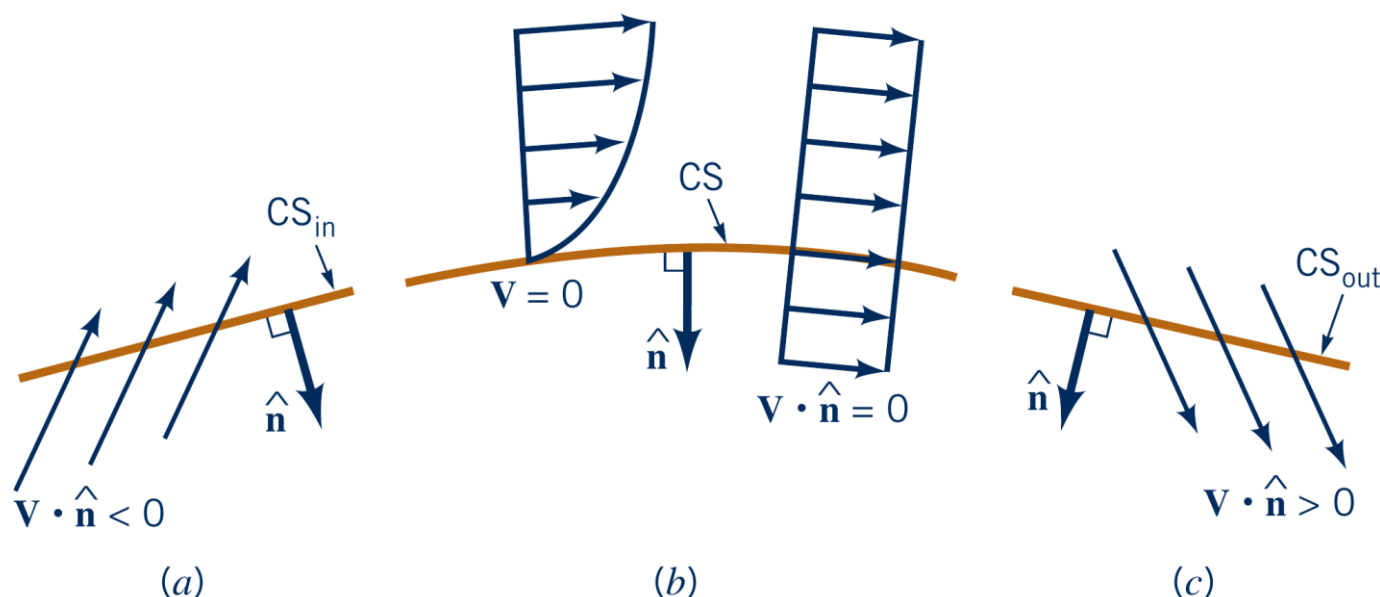
For an arbitrary fixed control volume, the 2 flat faces become the whole control surface and the sum becomes an integral.

Only the component of velocity normal to the surface carries fluid across it, so the sign takes care of itself:

$$\mathbf{v} \cdot \hat{n} = |\mathbf{v}| \cos \theta \quad \begin{cases} > 0 & \text{outflow} \\ = 0 & \text{no flow} \\ < 0 & \text{inflow} \end{cases}$$

With $\hat{n}$ the outward unit normal, the theorem becomes:

$$\frac{DB_{\text{sys}}}{Dt} = \frac{\partial}{\partial t} \int_{\text{cv}} \rho b \, d\mathcal{V} + \int_{\text{cs}} \rho b \, \mathbf{v} \cdot \hat{n} \, dA$$



Inflow, no flow and outflow on portions of the control surface.

Reynolds transport theorem

$$\underbrace{\frac{DB_{\text{sys}}}{Dt}}_{\text{system}} = \underbrace{\frac{\partial}{\partial t} \int_{\text{cv}} \rho b \, d\mathcal{V}}_{\text{unsteady}} + \underbrace{\int_{\text{cs}} \rho b \, \mathbf{v} \cdot \hat{n} \, dA}_{\text{convective}}$$

- **The left-hand side is Lagrangian:** it is the quantity the physical law actually refers to.
- The **first term** on the RHS is the unsteady effect — the amount of B inside the box may be changing.
- The **second term** is the convective effect — B is being carried across the boundary.

That is the same pair of effects as in the material derivative, written for a finite volume instead of a particle.

The theorem is the integral counterpart of $D(\cdot)/Dt$. For steady flow the unsteady term vanishes and

$$\frac{DB_{\text{sys}}}{Dt} = \int_{\text{cs}} \rho b \, \mathbf{v} \cdot \hat{n} \, dA$$

which is where the conservation of mass, the momentum equation and the energy equation all begin.

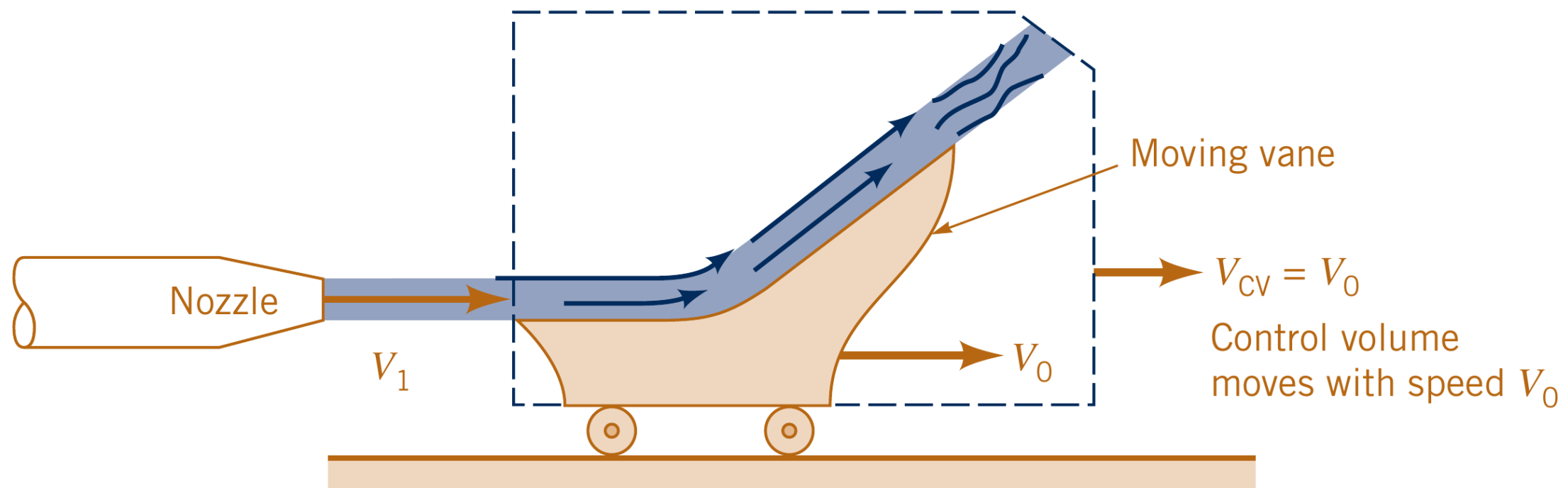
Moving control volumes

A control volume may translate with constant velocity. That is the natural choice when the fluid strikes a moving blade, as in a turbine.

What changes? only the velocity in the flux term: fluid crosses a moving surface at the relative velocity $\mathbf{w}$, not at $\mathbf{v}$:

$$\frac{DB_{\text{sys}}}{Dt} = \frac{\partial}{\partial t} \int_{\text{cv}} \rho b \, d\mathcal{V} + \int_{\text{cs}} \rho b \, \mathbf{w} \cdot \hat{\mathbf{n}} \, dA, \quad \mathbf{v} = \mathbf{w} + \mathbf{v}_{\text{cv}}$$

Example: A jet moving at $30 \, \text{m/s}$ strikes a vane moving at $6 \, \text{m/s}$ ahead of it; an observer riding on the vane sees the water arrive at $24 \, \text{m/s}$.

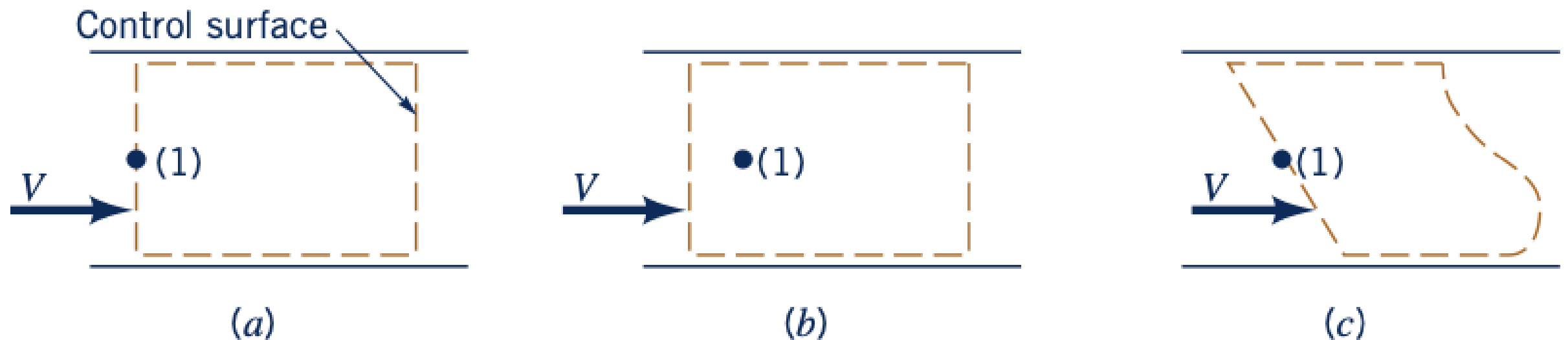


A jet striking a vane moving at v_0 ; the dashed control volume travels with it. Munson et al., Fig. 4.20

Selecting a control volume

Choosing a control volume is like choosing a free-body diagram: none is wrong, but some are far less work. Three rules cover most problems:

- 1. Put the unknown on the surface:** not buried inside, so that it appears in the flux integral.
- 2. Cut normal to the flow.** then θ is 0° or 180° and $\mathbf{v} \cdot \mathbf{n}$ is simply $\pm v$.
- 3. Let the rest follow a solid wall.** no fluid crosses it, $\mathbf{v} \cdot \mathbf{n} = 0$, so it contributes nothing.



Three control volumes for the same pipe flow. To find the pressure at point (1),

- (a) is the best choice: the point lies on the control surface, and the surface is normal to the flow.
- (b) In (b) the point is buried inside;
- (c) in (c) the surface cuts the flow obliquely.

Choosing a control volume

Gas escaping from a pressurised aerosol can.

- A control volume that followed the edge of the expanding cloud would be hopeless — the shape is complicated and is not known in advance.
- The fixed volume drawn with the dashed line is far better: its surface follows the can, so the surface integrals reduce to the small area where the gas actually leaves.



Key equations

Material derivative

$$\mathbf{a} = \frac{D\mathbf{v}}{Dt}, \quad \frac{D(\quad)}{Dt} \equiv \frac{\partial(\quad)}{\partial t} + (\mathbf{v} \cdot \nabla)(\quad)$$

The rate of change following the particle: a local term plus a convective term

Reynolds transport theorem

$$\frac{DB_{\text{sys}}}{Dt} = \frac{\partial}{\partial t} \int_{\text{cv}} \rho b \, d\mathcal{V} + \int_{\text{cs}} \rho b \, \mathbf{v} \cdot \hat{n} \, dA$$

The same two effects for a finite control volume: storage plus flux through the surface

Lecture key words

- Field representation: velocity and acceleration fields
- Eulerian vs. Lagrangian methods
- Steady and unsteady flows
- Streamline, Streakline and Pathline
- Material derivative
- Convective acceleration
- Extensive and intensive properties
- System vs. control volume
- Reynolds transport theorem